

Assignment 1: Student Risk Prediction using Multi-Layer Perceptron (MLP) Backpropagation (Partial Mid-term Assessment)

Course Title: Deep Learning

Submission Method: Online via the Moodle LMS

Course Code: CAI3103

Submission Due Date: Tuesday, April 07, 2026

Department: Artificial Intelligence

Submission Due Time: 11:30 PM

Lecturer name: Prof. Nashwa El-Bendary

Marks: 6

Assignment Details:

A university's Computer Science department wants to build a predictive model to identify at-risk students early in the semester. Given two standardized features: Study Hours (f_1) and Sleep Quality (f_2), with a target outcome ($t=1$ for Pass, $t=0$ for Fail). It's required to perform the MLP backpropagation algorithm for a sample dataset of two distinct student patterns (**Student A** (P1) and **Student B** (P2)), given in the table below:

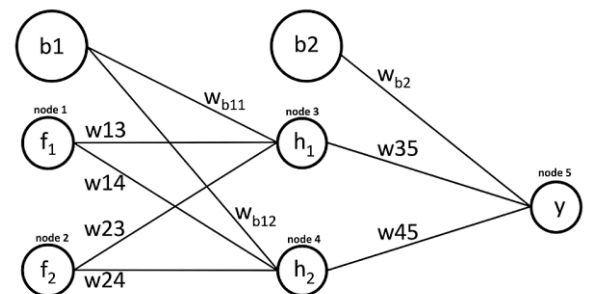
Pattern	f_1	f_2	t	w_{13}	w_{14}	w_{23}	w_{24}	w_{35}	w_{45}	w_{b11}	w_{b12}	w_{b2}
P1	0.8	0.6	1	0.4	0.2	0.3	0.5	0.7	0.7	0.1	0.4	0.3
P2	0.2	0.3	0									

Assume weights are updated **stochastically** (after each individual pattern).

ANN Network Architecture:

The model utilizes a standard three-layer feed-forward Artificial Neural Network (ANN) with the following characteristics (as shown in the figure below):

- **Input Layer:** Two feature nodes (f_1, f_2 ; as nodes 1 and 2) and a bias (b_1) to offset the hidden layer.
- **Hidden Layer:** Two hidden nodes (h_1, h_2 ; as nodes 3 and 4) using a weighted sum and Sigmoid activation for non-linearity.
- **Output Layer:** One output node (y ; node 5) that processes hidden layer signals and a second bias (b_2) to output the final pass probability.
- Assume bias inputs $b_1 = 1$ and $b_2 = 1$, learning rate $\eta = 0.5$, and Sigmoid activation function $f(a) = \frac{1}{1+e^{-a}}$ are used for all calculations.



Detailed Computational Requirements:

Show all detailed mathematical steps to demonstrate how the network weights adjust over **two full epochs** to minimize classification error. Your solution must include:

1. **Forward Propagation:** Calculate the input and activation values for hidden nodes 3 and 4, and output node 5.
2. **Error Computation:** Determine the error for each pattern based on the target value t .
3. **Backward Propagation:** Demonstrate the calculation of local gradients (δ) and the resulting weight updates (Δw) for all connections, including biases.
4. **Final Summary:** Provide a table of the final adjusted weights at the conclusion of each epoch.